

October 1, 2025 11:41 PM GMT

Global Macro Commentary | Global

October 1: The Shutdown Starts

US enters government shutdown; US rates bull-steepen on weak private payrolls and potential government layoffs; solid BoJ Tankan; Italy draft budget plan shows improving fiscals; India RBI remains on hold; US stocks gain; DXY at 97.706 (-0.1%); US 10y at 4.098 (-5.2bp).

Global front-end rates outperform as the US enters a government shutdown.

- **As the US enters a government shutdown, the administration reiterates proposals to implement permanent federal layoffs, suggesting that the move could come in the next 1-2 days.** While details of the plan remain unclear, the move is already facing [legal challenges](#). Meanwhile, an [estimated](#) 750,000 federal employees are furloughed, and the BLS will not release economic data until after the shutdown. On Tuesday evening, the US administration [withdrew](#) the nomination of E.J. Antoni for BLS commissioner, suggesting that Antoni did not have enough votes to be confirmed. The Supreme Court schedules a hearing regarding Fed Governor Cook for January 2026 and declines to allow her removal while the case proceeds, meaning that Governor Cook will likely remain in her role until 2026.
- **US rates bull-steepen on a downside surprise in private payrolls and potential concerns around the shutdown increasing unemployment,** particularly in light of the administration's proposed layoffs. ADP payrolls data comes in very weak at -32k against the 52k gain expected by consensus. Although the drop was partially driven by a 43k downward benchmark adjustment, it nevertheless portrays a weakening labor market. The ADP report has historically correlated poorly with the BLS employment report, but with the latter expected to be delayed by the shutdown, our economists believe this data boosts the case for near-term Fed cuts (for more, see [here](#)). Accordingly, front-end Treasuries rally 7bp, with the market-implied odds of an October Fed cut rising to 98%.
- **The dollar trades sideways (DXY: -0.1%) as US equities strengthen throughout the day (S&P 500: +0.3%) on constructive trade developments.** Healthcare stocks extend their rally from Tuesday, gaining another 3% as the US administration [delays](#) implementing the pharmaceutical tariffs originally slated to take effect today. The move is intended to provide more time for negotiations with pharmaceutical companies over lowering domestic drug prices, in line with our public policy strategists' [expectations](#). President Trump also [confirms](#) that he will meet with Chinese President Xi in four weeks, most likely at the October 31 APEC summit, in line with prior reporting.
- **In Japan, the BoJ Tankan shows solid confidence among Japanese manufacturers, but the market reaction is muted as the survey gives few signals on near-term monetary policy.** Super-long end JGBs sell off in the

MORGAN STANLEY & CO. LLC

Bradley Tian

Strategist

Bradley.Tian@morganstanley.com

+1 212 761-0826

Molly Nickolin

Strategist

Molly.Nickolin@morganstanley.com

+1 212 761-3592

Heather Berger

Economist

Heather.Berger@morganstanley.com

+1 212 761-2296

Sofia Palacios

Strategist

Sofia.Palacios@morganstanley.com

+1 212 761-0428

MORGAN STANLEY MUFG SECURITIES CO., LTD.+

Hiromu Uezato

Strategist

Hiromu.Uezato@morganstanleymufg.com

+81 3 6836-8431

MORGAN STANLEY ASIA LIMITED+

Gek Teng Khoo

Strategist

Gek.Teng.Khoo@morganstanley.com

+852 3963-0303

MORGAN STANLEY INDIA COMPANY PRIVATE LIMITED+

Nimish M Prabhune

Strategist

Nimish.Prabhune@morganstanley.com

+91 22 6996-1862

MORGAN STANLEY & CO. INTERNATIONAL PLC+

David Cueva

Strategist

David.Cueva@morganstanley.com

+44 20 7425-2143

Maria Chiara Russo

Strategist

Maria.Chiara.Russo@morganstanley.com

+44 20 7677-3499

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Tokyo morning in line with the overnight twist-steepening in US rates while front-end yields drift lower in the afternoon, resulting in a twist-steepening (2y: -0.5bp; 30y: +1bp). JPY outperforms the G10, closing the New York session 0.6% higher against USD as yields fall in markets outside Japan. We expect Thursday's 10y auction to be digested smoothly, but without meaningful follow-through given lingering risks from the LDP leadership election on Saturday and the ongoing US shutdown.

- **European rates twist steepen as euro area CPI comes in line with expectations at 2.2%**, despite upside [surprises](#) in Tuesday's individual country data. Front-end bund yields fall 1bp in sympathy with the US, but the long end lags the rally (20y: +2bp) amid speculative [reports](#) suggesting a potential increase in German supply. In Italy, the BTP curve slightly outperforms (2y: -2bp; 30y: 0bp) as Italy's budget draft plan confirms expectations for improving fiscals, with the 2025 deficit-to-GDP [reported](#) to be 3% — in line with the EU's target and below the 3.3% deficit expected back in April. The lower deficit could potentially allow Italy to exit the EU's excessive deficit procedure.
- **In a quiet session, LatAm rates largely move in line with the US.** Brazil's DI curve underperforms, with yields rising 5bp in the belly. In Colombia, COP outperforms (USD/COP: -1.0%) following BanRep's [decision](#) to maintain rates on Tuesday.
- **Brent oil slips 1% as the US EIA reports a 1.8mn barrel weekly increase in crude inventories**, with market participants continuing to anticipate an OPEC+ supply hike announcement this weekend. Lower oil prices and weakening Canadian PMI data (47.7; P: 48.3) lead CAD to weaken 0.1% against USD. Gold climbs another 0.2%, potentially supported by safe-haven demand amid the US shutdown.

Central Bank Monitor

Developed Markets

- **Chicago Fed President Goolsbee** [reiterated](#) concerns about the recent pickup in services inflation, which could mean price pressures are persistent in parts of the economy least impacted by tariffs. He stated that the lack of official data during a government shutdown will make it harder for central bankers to gauge the economy, particularly on the inflation side.
- **BoC September meeting minutes** [showed](#) that policymakers discussed maintaining policy rates due to improving consumption data before ultimately deciding to lower rates. The BoC noted that uncertainty surrounding the status of the USMCA trade agreement would likely impede business investment.

Emerging Markets

- In **India**, the **RBI** maintained its policy rate at 5.50% at its October meeting (MSe: 5.25%; C: 5.50%; P: 5.50%) in a unanimous decision. The RBI [noted](#) the uncertain impact of front-loaded monetary policy actions and recent fiscal measures as factors supporting a patient approach towards further action.

The RBI revised its FY26 GDP growth forecast to 6.8% y/y (P: 6.8%). The central bank maintained a neutral stance, noting that buoyant services, stable employment, and GST rate rationalization have supported demand. However, global headwinds such as tariff and trade policy uncertainties, geopolitical tensions, and market volatility pose risks to external demand and growth. On inflation, the RBI lowered its FY26 headline CPI projection to 2.6% y/y (P: 3.1%). The benign trend in near-term food inflation is supported by a favorable base effect, the southwest monsoon, ample reservoir levels, and sound kharif sowing.

- **RBI Governor Malhotra** stated that despite robust domestic fundamentals, INR has seen some depreciation accompanied by phases of volatility, adding that RBI is keeping a close watch and will take appropriate steps if needed. On monetary policy transmission, Malhotra noted that the 10y G-Sec yield does not respond one-for-one to policy rate changes, but he sees room for 10y yields to move lower.

Exhibit 1: G4 Rates Closes (5pm NY)

Tenor	US			Germany			UK			Japan		
	1-Oct	1d Δ (bp)	2wk	1-Oct	1d Δ (bp)	2wk	1-Oct	1d Δ (bp)	2wk	1-Oct	1d Δ (bp)	2wk
2y	3.535	(7.4)		2.010	(0.8)		3.961	(1.9)		0.952	1.4	
3y	3.546	(7.3)		2.086	(0.6)		3.960	(2.3)		1.006	0.6	
5y	3.672	(6.9)		2.299	(0.9)		4.115	(1.9)		1.233	1.1	
7y	3.867	(6.2)		2.443	(0.4)		4.279	(0.8)		1.473	1.2	
10y	4.098	(5.2)		2.712	0.2		4.695	(0.4)		1.646	0.4	
20y	4.672	(3.1)		3.198	1.2		5.390	0.3		2.613	1.4	
30y	4.710	(2.1)		3.298	1.8		5.510	0.4		3.152	1.5	

Source: Morgan Stanley Research, Bloomberg

Exhibit 2: Inflation Closes (5pm NY)

US TIPS				10y Real Yields, BEI							
Tenor	1-Oct	1d Δ (bp)	2wk	10y	1-Oct	1d Δ (bp)	2wk	10y	1-Oct	1d Δ (bp)	2wk
5y	1.217	(4.8)		DBRi	0.749	(1.1)		BTPi	1.854	(1.1)	
BEI	2.428	(2.4)		BEI	1.750	0.8		BEI	1.719	0.5	
10y	1.747	(3.0)		UKTi	1.674	1.0					
BEI	2.349	(2.1)		BEI	3.096	(1.3)					
30y	2.473	(0.8)		JGBi	0.028	(1.4)					
BEI	2.239	(1.3)		BEI	1.607	2.2					

Source: Morgan Stanley Research, Bloomberg

Exhibit 3: Macro Closes (5pm NY)

Spot	Majors			Spot	EM			Index	Equities / Commodities		
	1-Oct	1d Δ (%)	2wk		1-Oct	1d Δ (%)	2wk		1-Oct	1d Δ (%)	2wk
DXY	97.71	(0.1)		CNH	7.13	(0.0)		S&P	6711.20	0.3	
EUR	1.173	(0.0)		INR	88.69	(0.1)		Stoxx	5581.21	0.9	
GBP	1.348	0.2		THB	32.36	(0.1)		FTSE	9446.43	1.0	
JPY	147.07	(0.6)		KRW	1403.10	(0.0)		Nikkei	44550.85	(0.8)	
CHF	0.797	0.1		ZAR	17.23	(0.2)		VIX	16.29	0.1	
AUD	0.661	0.0		MXN	18.38	0.4		Gold (S)	3865.74	0.2	
CAD	1.394	0.1		BRL	5.33	0.1		WTI (Fut)	61.78	(0.9)	

Source: Morgan Stanley Research, Bloomberg

Central Bank OIS															
Fed	Implied	1d Δ (bp)	Total Δ	ECB	Implied	1d Δ (bp)	Total Δ	BDE	Implied	1d Δ (bp)	Total Δ	BOJ	Implied	1d Δ (bp)	Total Δ
EFFR (9/30)	4.990	0.0	—	EDONIA (9/30)	1.916	0.0	—	SONIA (9/30)	3.967	0.0	—	TOMAR (9/30)	0.477	0.0	—
10/29/25	4.082	(1.0)	(24.8)	10/30/25	1.925	0.0	0.9	11/06/25	3.944	(0.2)	(2.3)	10/30/25	0.635	(1.3)	15.8
12/10/25	3.622	(3.6)	(46.8)	12/18/25	1.906	0.2	(1.0)	12/18/25	3.908	(0.1)	(5.9)	12/19/25	0.677	0.0	20.0
01/28/26	3.507	(5.5)	(58.3)	02/05/26	1.893	0.1	(2.3)	02/05/26	3.804	(0.5)	(16.3)	01/23/26	0.716	(0.8)	23.9
03/18/26	3.386	(7.8)	(70.4)	03/19/26	1.855	0.2	(6.1)	03/19/26	3.739	(3.5)	(22.8)				
05/29/26	3.320	(8.0)	(77.1)	04/30/26	1.848	(0.3)	(6.6)	04/30/26	3.676	(1.2)	(29.2)				
06/17/26	3.188	(7.7)	(99.2)	06/11/26	1.834	(0.3)	(8.2)	06/18/26	3.651	(0.7)	(29.5)				

The Day Ahead

- 8:30am CET - **Switzerland September CPI** (C: 0.3% y/y, -0.2% m/m; P: 0.2% y/y, -0.1% m/m)
- 10:00am BST - **Euro Area August Unemployment Rate** (MSe: 6.2%; C: 6.2%; P: 6.2%)
- 5:30pm BST - **ECB Governing Council Member Villeroy** will speak in Luxembourg
- 7:30am EST - **US September Challenger Job Cuts** (MSe: -0.2% y/y; P: 13.3%)
- *Likely delayed by government shutdown*: US September Initial Jobless Claims (MSe: 220k ; C: 225k; P: 218k) & August Factory Orders (MSe: 1.4% ; C: 1.4%; P: -1.3%)
- 10:30am EST - **Dallas Fed President Logan** speaks at the University of Texas

- Thu 7:00am HKT - **Korea September Headline CPI** (MSe: 1.9% y/y; C: 2.0%; P: 1.7%); **Core CPI** (MSe: 1.9% y/y; C: 2.0%; P: 1.3%) & **Current Account Balance** (P: \$10.8bn)
- 4:30pm HKT - **Hong Kong August Retail Sales Value** (C: 2.0% y/y; P: 1.8%) & **Volume** (C: 1.9% y/y; P: 1.0%)
- 9:00pm HKT - **Singapore September Purchasing Managers Index** (P: 50.0)

Developed Markets

- **US September ISM Manufacturing:** 49.1 (MSe: 48.50; C: 49; P: 48.7). Manufacturing contracted for the seventh straight month in September, but production and employment improved slightly. New orders weakened to 48.9 from 51.4, and the prices index fell to 61.9 from 63.7, signaling softer demand despite a modest uptick in factory activity.
- **US September ADP Employment Change:** -32.0k (C: 50.5k; P: 54k). ADP private payrolls fell 32k in September. The decline was exacerbated by a re-benchmarking to QCEW tax data that contributed -43k. With ongoing deterioration and the absence of the BLS payrolls data, the negative print keeps the Fed on alert. [Read more](#)
- **Japan September BoJ Tankan Large Manufacturing Current DI:** +14pt (MSe: +15pt, C: +14pt, P: +13pt); **Large Manufacturing Outlook DI:** +12pt (MSe: +12pt, C: +13pt, P: +12pt); **Large Non-Manufacturing Current DI:** +34pt (MSe: +31pt, C: +34pt, P: +34pt), **Large Non-Manufacturing Outlook DI:**

+28pt (MSe: +30pt, C: +28pt, P: +27pt). Our economists see the outlook for business sentiment and earnings for 2H25 remaining generally cautious, reflecting concerns over tariffs. They maintain their view that there is no urgency for a rate hike in October. [Read more](#)

- **UK September Final PMI Manufacturing:** 46.2 (MSe: 46.2; C: 46.2; P: 46.2). In line with flash estimate.
- **Switzerland August Retail Sales:** 0.2% (P: 0.7%)
- **Switzerland September Manufacturing PMI:** 48.0 (C: 48.0; P: 49.0)
- **Euro Area September CPI:** 2.2% (MSe: 2.3%; C: 2.2%; P: 2.0%)
- **Netherlands September CPI:** 3.3% (MSe: 3.1%; P: 2.8%)
- **Canada September S&P Manufacturing PMI:** 47.7 (P: 48.3). The [survey](#) showed output, new orders, and exports all fell, likely causing employment reductions.

Emerging Markets

- **Chile August Economic Activity:** 0.5% y/y (P: 1.8%; C: 2.0%). Activity came significantly below consensus. The downside surprise was likely driven by the sharp contraction in mining output weighing heavily on the headline figure.
- **Mexico August Remittances Total:** \$5578.2 (P: \$5329.9m; C: \$5420.0m) An upside surprise, but the headline figure masks underlying weakness — total inflows fell 8.3% y/y, average remittance size declined 1.2% y/y to \$402, and the number of transactions dropped 7.2% m/m to 13.87 million. The data suggests that while the monthly print beat expectations, the broader trend remains soft.
- **Korea September Trade Balance:** \$9.6bn (MSe: \$8.5bn; C: \$8.0bn; P: \$6.5bn); **Exports:** 12.7% y/y (MSe: 9.6%; C: 7.8%; P: 1.2%) & **Imports:** 8.2% y/y (MSe: 6.7%; C: 5.6%; P: -4.1%)
- **Indonesia September Headline CPI:** 2.7% y/y (C: 2.5%; P: 2.3%); **September Core CPI:** 2.2% y/y (C: 2.2%; P: 2.2%)
- **Indonesia August Trade Balance:** \$5.5bn (C: \$4.1bn; P: \$4.2bn); **Exports:** 5.8% y/y (C: 6.5%; P: 9.9%) & **Imports:** -6.6% y/y (C: 0.2%; P: -5.9%)
- **India HSBC Final September Manufacturing PMI:** 57.7 (Preliminary: 58.5; P: 59.3)
- **Indonesia September S&P Manufacturing PMI:** 50.4 (P: 51.5)
- **Korea September S&P Manufacturing PMI:** 50.7 (P: 48.3)
- **Malaysia September S&P Manufacturing PMI:** 49.8 (P: 49.9)
- **Philippines September S&P Manufacturing PMI:** 49.9 (P: 50.8)
- **Thailand September S&P Manufacturing PMI:** 54.6 (P: 52.7)
- **Taiwan September S&P Manufacturing PMI:** 46.8 (P: 47.4)

For more details, please refer to our economists' weekly reports and data previews:

- [US](#)
- [Japan](#)
- [Europe](#)
- [Asia/Pacific](#)
- [LatAm](#)

- [CEEMEA](#)

Please [click here](#) if you would like to receive our weekly strategic advice on global rates, FX, and EM. See our latest [Global Macro Strategist: As the Grip of Tariffs Tightens](#) and recent publications:

- G10 FX Strategy: [AI Themes - Still a Sideshow for G10 FX](#)
- India Economics: [RBI Policy Review - Door for Easing Remains Ajar](#)
- Cross-Asset Playbook: [Speaking of Rich...](#)

G4 Smarter (Beta) Trading Partners

MORGAN STANLEY & CO. LLC

Matthew Hornbach, CMTMatthew.Hornbach@morganstanley.com

+1 212 761-1837

Since 2000, enhancements to a G4 10y government bond futures momentum strategy have produced higher Sharpe ratios and stronger returns relative to total return government bond indices for the G4: US, Germany, Japan, and the UK. See [A "Smarter" \(Beta\) Way to Trade G4 10y Futures Duration?](#) for more information on these strategies.

Exhibit 5: Trading Signals for G4 Smarter (beta) Trading Strategy

Current Risk, G4 10y Futures	G4 Strategy Weight	Trade Longs Portfolio	Fade Shorts Portfolio	Total Risk Trade Longs Only	Total Risk Trade Longs/Fade Shorts (max 200%)	Trade Longs Portfolio Entry Date	Trade Longs Portfolio Exit Date	Fade Shorts Portfolio Entry Date	Fade Shorts Portfolio Exit Date
J8 10y Future	32.50%	100%	100%	100%	200%	9/9/2025	10/17/2025	9/19/2025	10/28/2025
GE 10y Future	29.25%	100%	100%	100%	200%	9/8/2025	10/14/2025	9/15/2025	10/20/2025
US 10y Future	30.50%	0%	100%	0%	100%	-	-	9/24/2025	10/30/2025
UK 10y Future	7.75%	100%	100%	100%	200%	9/10/2025	10/15/2025	9/25/2025	10/30/2025

Source: Morgan Stanley Research

Trading Strategy 1 – "Trade Longs/Fade Shorts"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, buy the futures contract and hold for a 25-business-day period. In short, this strategy buys futures when the Simple Moving Average Crossover (SMAX) generates both a long and a short signal, given the historical outperformance of long signals traded long and the underperformance of short signals traded short. Given that the SMAX could generate both a long and a short signal within the predefined holding period, an investor may have a 200% long position since each of the two signals would be traded in separate portfolio sleeves.

Trading Strategy 2 – Trade "Longs Only"

When the 5-day moving average crosses above the 20-day moving average, buy the futures contract (long duration) and hold for a 25-business-day period. When the 5-day moving average crosses below the 20-day moving average, do nothing. In short, an investor *only* trades long signals initiated by the SMAX given their historical precedent to outperform.

US Holiday Trading Strategy

Currently Active Strategies

None.

Combining the individual trading signals based on 10y Treasury future (TY) price patterns before and after US holidays has generated an annual strategy with no down years since 1987. See [Trading 10y Treasury Futures Around US Holidays](#) for more information.

Using historical data since 1983, we created a trading strategy using 10y Treasury futures (TY) that trades 11 times per year, either long or short, before or after the following US holidays: New Year's Day, Martin Luther King Day, Presidents' Day, Good Friday, Memorial Day, Independence Day, Labor Day, Columbus Day, Thanksgiving, and Christmas. [Exhibit 6](#) displays the trading rules:

Exhibit 6: Trading rules for US holiday trading model using US 10y futures contracts (TY)

Holiday	Day of Week	Direction	Before/After Holiday	Trading Rule (business days, 3pm NY unless specified)	Trade Entry (business days)	Trade Exit (business days)
New Year's Day	Rotates	Long TY	After	Buy TY New Years +1, Sell TY New Years +4	New Year + 1	New Year + 4
Martin Luther King Day 1	Monday	Long TY	Before	Buy TY Tuesday before MLK, Sell Friday before MLK	MLK - 4	MLK - 1
Martin Luther King Day 2	Monday	Short TY	After	Sell TY Tuesday after MLK, Buy Friday after MLK	MLK + 1	MLK + 4
President's Day	Monday	Long TY	After	Buy TY Tuesday after Pres. Day, Sell TY Friday after Pres. Day	Pres. Day + 1	Pres. Day + 4
Good Friday	Friday	Long TY	After	Buy TY Monday after Good Friday, Sell TY Friday after Good Friday	Good Friday + 1	Good Friday + 5
Memorial Day	Monday	Long TY	After	Buy TY Tuesday after Mem. Day, Sell TY Friday after Mem. Day	Mem. Day + 1	Mem. Day + 4
Independence Day (July 4th)	Rotates	Long TY	After	Buy TY Independence Day + 2, Sell TY Independence Day + 4	Ind. Day + 2	Ind. Day + 4
Labor Day	Monday	Long TY	Before	Buy TY Monday before Labor Day, Sell TY Thursday before Labor Day	Labor Day - 4	Labor Day - 2
Columbus Day	Monday	Short TY	Before	Sell TY Monday before Col. Day, Buy TY Thursday before Col. Day	Colum. Day - 5	Colum. Day - 2
Thanksgiving	Thursday	Long TY	Before	Buy TY Friday before Thanksgiving, Sell TY Tuesday before Thanksgiving	Thanksgiving - 4	Thanksgiving - 2
Christmas	Rotates	Short TY	Before	Sell TY Christmas Day - 4, Buy TY Christmas Day - 1	Christmas - 4	Christmas - 1

Source: Morgan Stanley Research. Note: Past performance is no guarantee of future results.

Global Macro Strategy Team

MORGAN STANLEY & CO. LLC	Matthew Hornbach, CMT Matthew.Hornbach@morganstanley.com	Global Head of Macro Strategy	+1 212 761-1837
	Bradley Tian	Global Macro Commentary	+1 212 761-0826
	Martin Tobias, CFA, CMT	US Rates Strategist	+1 212 761-6076
	Shaun Zhou	US Rates Strategist	+1 212 761-3348
	Aryaman Singh	US Rates Strategist	+1 212 761-1993
	Eli Carter	US Rates Strategist	+1 212 761-4703
	Andrew Watrous	G10 FX Strategist	+1 212 761-5287
	Zoe K. Strauss	G10 FX Strategist	+1 212 761-0407
	Molly Nickolin	G10 FX Strategist	+1 212 761-3592
	Simon Waever Simon.Waever@morganstanley.com	Head of EM Sovereign Credit and Latin America Fixed Income Strategy	+1 212 296-8101
MORGAN STANLEY & CO. INTERNATIONAL PLC	Ioana Zamfir	Latin America Macro Strategist	+1 212 761-4012
	Emma Cerda	Latin America Sovereign Credit	+1 212 761-2344
	Sofia Palacios	Latin America Macro Strategist	+1 212 761-0428
	James K. Lord James.Lord@morganstanley.com	Global Head of FXEM Strategy	+44 20 7677-3254
	Gianluca Salford Luca.Salford@morganstanley.com	Head of European Rates Strategy	+44 20 7677-1337
	Lorenzo Testa	Euro Area Rates Strategist	+44 20 7677-0337
	Fabio Bassanin, CFA	UK Rates Strategist	+44 20 7425-1869
	Maria Chiara Russo	Euro Area Rates Strategist	+44 20 7677-3499
	David S. Adams, CFA David.S.Adams@morganstanley.com	Head of G10 FX Strategy	+44 20 7425-3518
	Neville Mandimika	CEEMEA Sovereign Credit Strategist	+44 20 7425-2509
MORGAN STANLEY ASIA LIMITED+	Arnav Gupta	CEEMEA Rates Strategist	+44 20 7677-0382
	David Cueva Oemer	CEEMEA Rates Strategist	+44 20 7425-2143
MORGAN STANLEY MUFG SECURITIES CO., LTD.	Gek Teng Khoo	AXJ Macro Strategist	+852 3963-0303
MORGAN STANLEY MUFG SECURITIES CO., LTD.	Koichi Sugisaki Koichi.Sugisaki@morganstanleymufg.com	Head of Japan Macro Strategy	+81 3 6836-8428
	Hiromu Uezato	Japan Macro Strategist	+81 3 6836-8431

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley Canada Limited and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFG Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/researchdisclosures, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: David S. Adams, CFA; Fabio Bassanin, CFA; Eli P. Carter; David Cueva; Arnab Gupta; Matthew Hornbach; Gek Teng Khoo; James K. Lord; Molly Nickolin; Sofia Palacios; Nimish M. Prabhune; Maria Chiara Russo; Luca Salford; Aryaman Singh; Zoe K. Strauss; Koichi Sugisaki; Lorenzo Testa; Bradley Tian; Martin W. Tobias, CFA; Hiromu Uezato; Andrew M. Watrous; Ioana Zamfir.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflict/policies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Germany.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Germany.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Germany, Japan, United Kingdom, United States of America.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Germany.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Germany, Japan, United Kingdom, United States of America.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Germany.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2025)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1499	41%	389	46%	26%	702	41%
Equal-weight/Hold	1618	44%	375	44%	23%	782	45%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	577	16%	88	10%	15%	234	14%
Total	3,698		853			1719	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding the relationship between the companies that are the subject of Morgan Stanley Research and Morgan Stanley Smith Barney LLC or Morgan Stanley or any of their affiliates, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to www.morganstanley.com/researchdisclosures.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley & Co. International PLC and its affiliates have a significant financial interest in the debt securities of Germany, Japan, United Kingdom.

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of

setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section

21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

The following authors are neither Equity Research Analysts/Strategists nor Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity or fixed income securities: Heather Berger.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Molly Nickolin; David S. Adams, CFA; Eli P Carter; Koichi Sugisaki; James K Lord; Aryaman Singh; Zoe K Strauss; Bradley Tian; Matthew Hornbach; Nimish M Prabhune; Ioana Zamfir; Gek Teng Khoo; Martin W Tobias, CFA; Andrew M Watrous; Lorenzo Testa; Luca Salford; Fabio Bassanin, CFA; Maria Chiara Russo; Hiromu Uezato; David Cueva.